

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

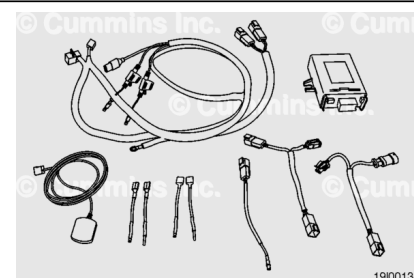
- INLINE™ 7 Data Link Adapter Kit, Part Number 5299899
- Data link adapter for split network, Part Number 5394863
- INSITE™ Electronic Service Tool Software Kit, Part Number 3886388
- Cummins Guidanz™ electronic service tool
- Multimeter, Part Number 3400162, or equivalent
- Electrical Test Lead Kit, Part Number 5299367

Additional Service Items

- No additional service items required.

General Information

Note : See the following link to download the latest version of the Cummins Guidanz™ electronic service tool. <https://www.cummins.com/support/electronic-service-tools-support/guidanz-support> (<https://www.cummins.com/support/electronic-service-tools-support/guidanz-support>)



The Cummins® route parameter manager will facilitate the addition of several new fuel economy features, including Predictive Cruise Control.

Cummins Inc. offers two solutions for route parameter manager installation:

- First fit installation is installed by the original equipment manufacturer (OEM) during the initial vehicle build.
- An Aftermarket Route Parameter Manager Kit is available as a retrofit option for most vehicle installations. This kit is sold and installed by Cummins® Authorized Repair Locations.

During first fit installation, Cummins Inc. provides **only** the route parameter manager. The OEM is responsible for the following:

- Harness
- Global Positioning System (GPS) antenna
- Installation
- All associated hardware.

The Route Parameter Manager Aftermarket Kit consists of the following:

Part Description	Part Number
Route Parameter Manager Module	3694010
Wiring Harness	5394153
Calibration Cable	5394154
Antenna	5394155
Electrical Connector Adapter	5394177
Electrical Connector Adapter	5394178
Electrical Connector Adapter	5394179
Wiring Harness Kit	5405054
Electrical Wire	5394173
Electrical Wire	5394174
Electrical Wire	5394175
Electrical Wire	5394176
Hook Loop Strip	5394181

The Route Parameter Manager Kit can be installed in all Cummins® Authorized Repair Locations.

Identifying First Fit Installation and Aftermarket Installation

- The Aftermarket Route Parameter Manager Kit harness is RED in color.
- The Aftermarket Route Parameter Manager Kit harness will have a Cummins® Part Number sticker located at the Universal Serial Bus (USB) and power branch of the harness.
- This can be verified at the location of the fuse panel.

Identifying first fit installation versus aftermarket installation is critical for warranty processing.

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

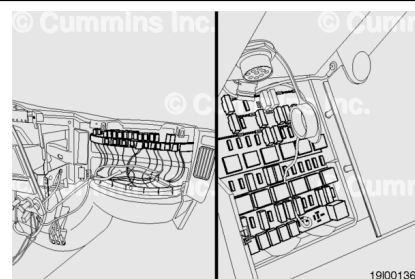
- Disconnect the batteries. See equipment manufacturer service information.

Remove

Route Parameter Manager Module

Locate the cab fuse panel.

Note : Fuse panel location varies by OEM. See equipment manufacturer service information for fuse panel location.



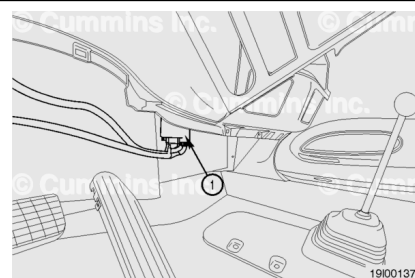
Follow the red route parameter manager harness to the module (1).

Remove all decorative covers while tracking the harness to the module. See equipment manufacturer service information.

Locate the route parameter manager module.

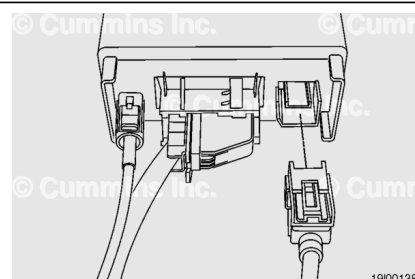
Note : The most common location is in the center of the console.

Note : For detailed information on route parameter module location, see Installation Instructions, Bulletin 5414612.



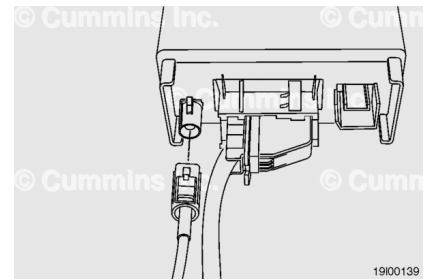
Push down on the locking tab of the USB connector.

Disconnect the USB cable connector from the route parameter manager module.



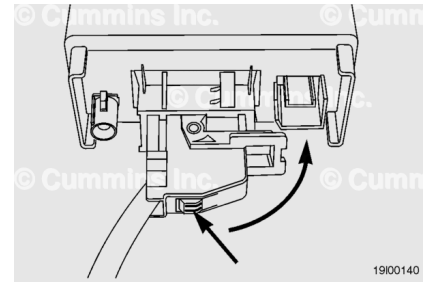
Push down on the locking tab of the blue GPS antenna connector.

Disconnect the blue GPS antenna connector from the blue plug on the route parameter manager module.



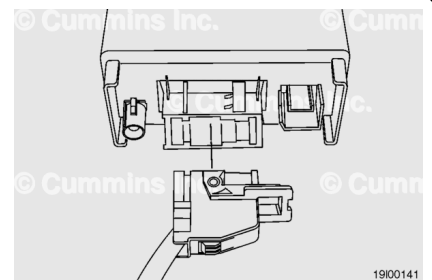
Disconnect the harness connector from the module.

Push down on the locking tab. Pull the lever past the locking tab, toward the module.



Pull the black harness connector away from the route parameter manager module.

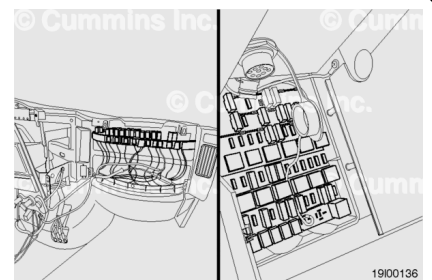
Remove the route parameter manager module from the cab.



Route Parameter Manager Harness

Locate the cab fuse panel.

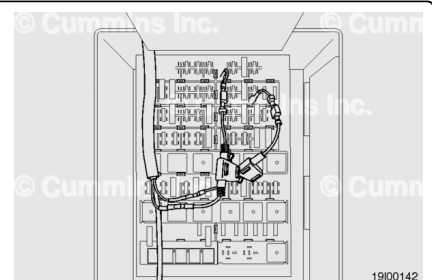
Note : Fuse panel location varies by OEM. See equipment manufacturer service information for fuse panel location.



Locate and remove the battery blade fuse adapter, the adapter made with RED wires, from the fuse panel.

Locate and remove the Key-ON blade fuse adapter, the adapter made with BLUE wire, from the fuse panel.

Note : Both wires will be connected to the RED route parameter manager harness.



Verify that the USB connector is not connected to any parts of the fuse panel. Remove zip ties or other mounting hardware used to secure USB connector to the fuse panel.

Follow the red route parameter manager harness to the module.

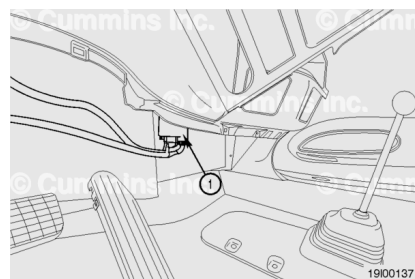
Remove all decorative covers while tracking the harness to the module. See equipment manufacturer service information.

Remove zip ties or other mounting hardware used to secure the route parameter manager harness in the cab.

Locate the route parameter manager module.

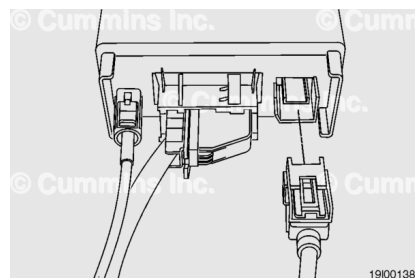
Note : The most common location for the route parameter manager module is in the center of the console.

Note : For detailed information on route parameter module location, see Installation Instructions, Bulletin 5414612.



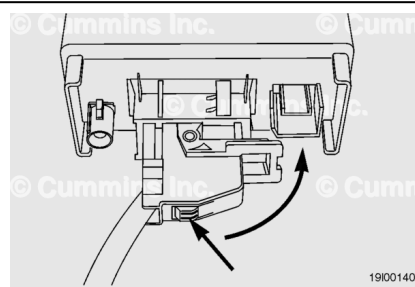
Push down on the locking tab of the USB connector.

Disconnect the USB cable connector from the route parameter manager module.



Disconnect the harness connector from the module.

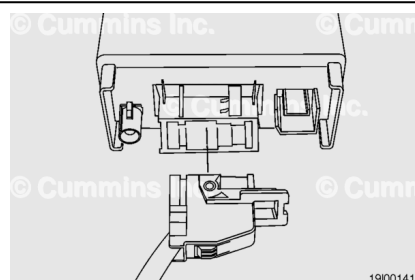
Push down on the locking tab. Pull the lever past the locking tab, toward the module.



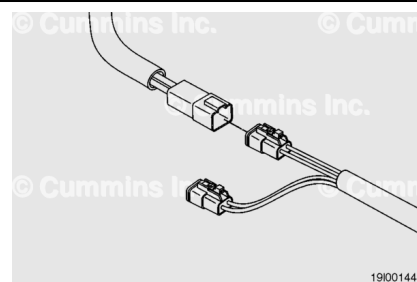
Pull the black harness connector away from the route parameter manager module.

Continue to follow the red route parameter manager harness to the terminating resistor connector location.

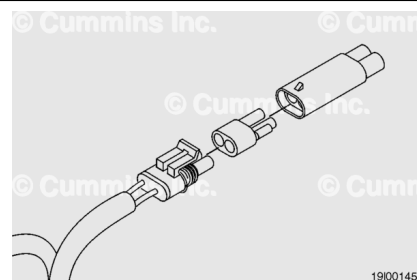
Remove zip ties or other mounting hardware used to secure route parameter manager harness in the cab.



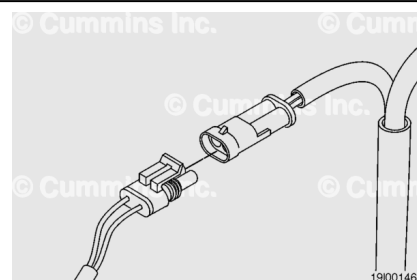
Disconnect the Y-adapter from the route parameter manager harness.



Remove the terminating resistor from the Y-adapter.



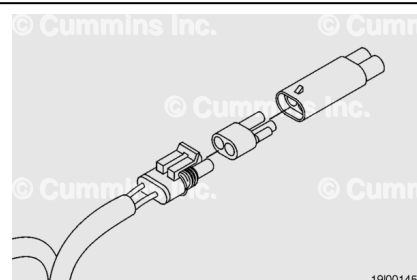
Remove the Y-adapter from the OEM harness.



Relocate the terminating resistor to the OEM harness.

Remove the Y-adapter from the cab.

Remove the route parameter manager harness from the cab.



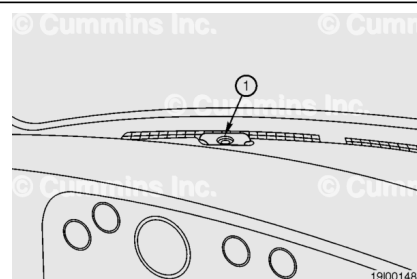
Route Parameter Manager Module GPS Antenna

Locate the route parameter manager module GPS antenna (1). It should be located on top of the dashboard.

Follow the GPS antenna wire to the module.

Remove all decorative covers while tracing the wire to the module. See equipment manufacturer service information.

Remove zip ties or other mounting hardware used to secure the GPS antenna wire in the cab.



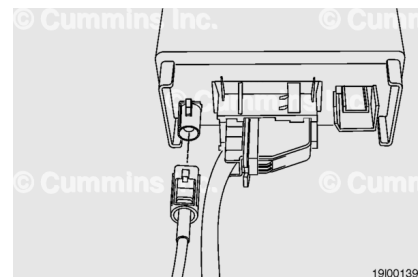
Note : The most common location for the route parameter manager module is in the center of the console.

Note : For detailed information on route parameter manager module location, see Installation Instructions, Bulletin 5414612.

Push down on the locking tab of the blue GPS antenna connector.

Disconnect the blue GPS antenna connector from the blue plug on the route parameter manager module.

Remove the GPS antenna from the cab.



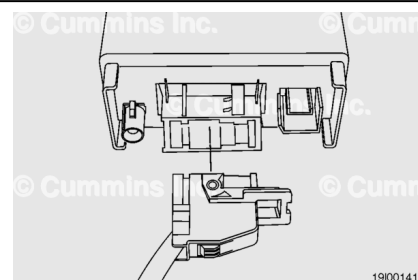
Install

Route Parameter Manager Module

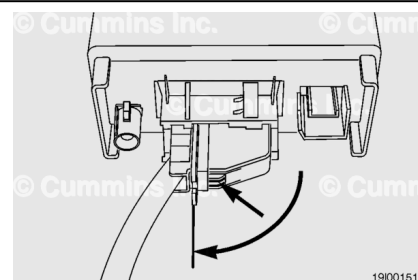
Mount the route parameter manager module to the cab structure. Use double-sided tape or wire ties.

Make sure the module has appropriate support and will **not** become loose over time.

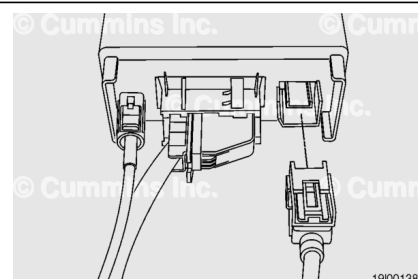
Connect the square, black connector of the harness to the route parameter manager module.



As the connector is inserted and snaps past the locking tab, make sure the locking lever is turned 90 degrees.



Connect the USB cable connector of the harness to the route parameter manager module.



Connect the blue GPS antenna connector to the blue plug on the route parameter manager module.

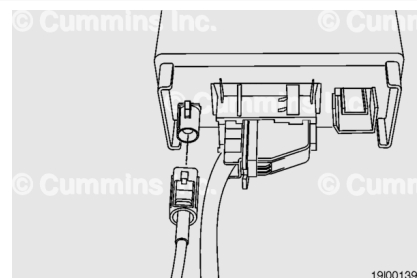
Make sure it has fully snapped into place.

To complete installation:

- Connect the batteries. See equipment manufacturer service information.
- Verify the route parameter manager module software version. Update if needed. Refer to Procedure 019-647 in Section 19. (/qs3/pubsys2/xml/en/procedures/377/377-019-647.html)
- Verify the route parameter manager GPS map version. Updated if needed. Refer to Procedure 019-648 in Section 19. (/qs3/pubsys2/xml/en/procedures/377/377-019-648.html)
- Check for active or inactive fault codes.

Note : Map and route parameter manager module software updates are required only if identified as a warrantable requirement in the engine control module (ECM) Calibration Revision History in QuickServe® Online.

Install all cab decorative covers. See equipment manufacturer service information.



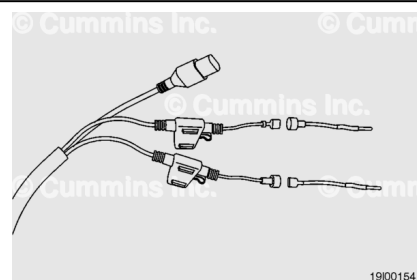
Route Parameter Manager Harness

Install the wiring harness.

Connect the battery fuse adapter to the route parameter manager harness.

Connect the Key-ON fuse adapter to the route parameter manager harness.

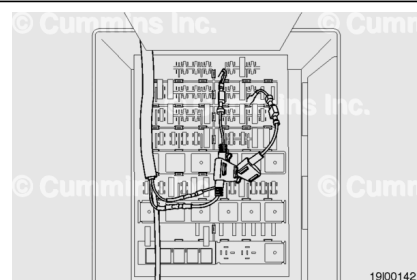
Note : The battery fuse adapter is RED in color. The Key-ON fuse adapter is BLUE in color.



Install the battery fuse adapter and Key-ON fuse adapter.

Mount the map update USB connector next to the cab fuse panel.

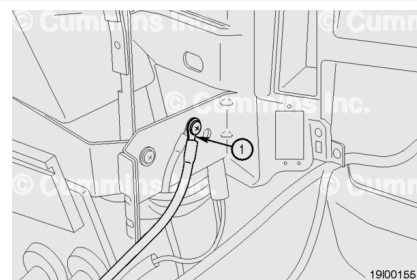
Note : The use of a multimeter can be required to verify power availability at the selected fuse location and powered side of the fuse.



Note : Only one side of the fuse opening will have power. Insert the fuse adapter into the powered side of the fuse mounting location.

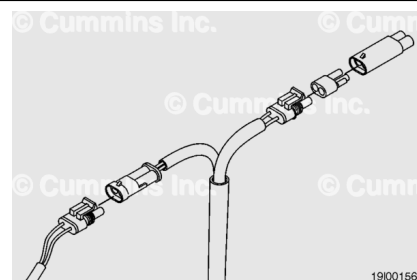
Connect the ground wire.

Note : If the eyelet terminal (ground lug) (1) is too small for proper assembly, replace the eyelet terminal with one of appropriate size.



Connect the Y-adapter to the OEM harness.

Install the terminating resistor on the Y-adapter.



Connect the Y-adapter to the route parameter manager harness data link connector.

There are two connectors on the route parameter manager harness:

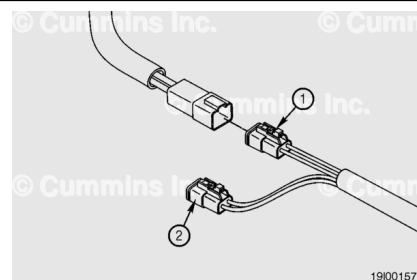
- 250k baud connector for all 2013 to 2015 engines, and for 2016 and newer engines with a split data link network (1)
- 500k baud connector for 2016 and newer engines with 500k baud network **only** (2).

Note : For more information on the data link selection, see the Route Parameter Manager Aftermarket Kit installation instruction.

Note : Failure to select correct data link network speed will cause Fault Codes 6575 and 6753 to become active.

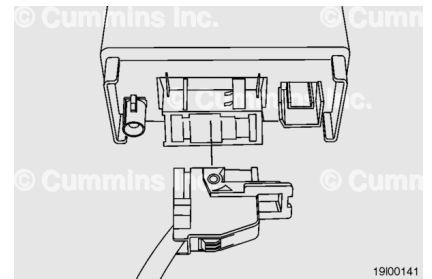
Secure the route parameter manager harness to the cab and dash substructure. Use zip ties.

Note : If there is extra wire left due to routing, coil it and tack neatly away to avoid wire damage.

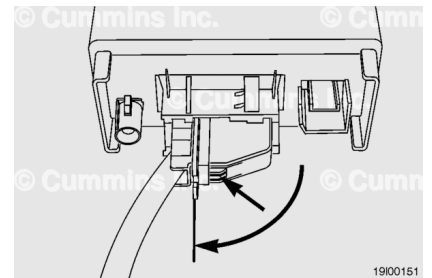


Connect the square, black connector of the harness to the route parameter manager module.

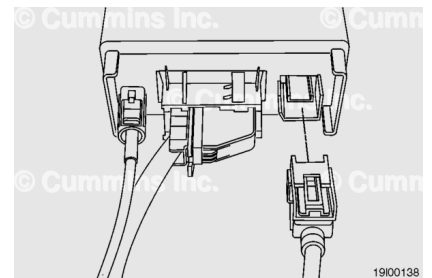
Make sure the locking lever is turned 90 degrees as the connector is inserted and snaps past the locking tab.



Make sure the locking lever is turned 90 degrees as the connector is inserted and snaps past the locking tab.



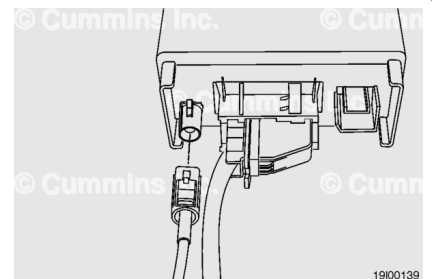
Connect the USB cable connector of the harness to the route parameter manager module.



Connect the blue GPS antenna connector to the blue plug on the route parameter manager module.

Make sure it has fully snapped into place.

Install all cab decorative covers. See equipment manufacturer service information.



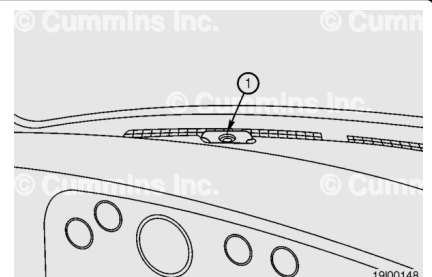
Route Parameter Manager Module GPS Antenna

Clean the mounting surface. Use the alcohol swab included in the kit.

Attach the GPS antenna to the dashboard. Use double-sided tape. The GPS antenna cable **must** point down once the antenna is mounted.

Note : The GPS antenna should be located inside the cab near the windshield so it gets good satellite signal.

Note : Failure to mount the GPS antenna will cause incorrect position readings for the route parameter



manager.

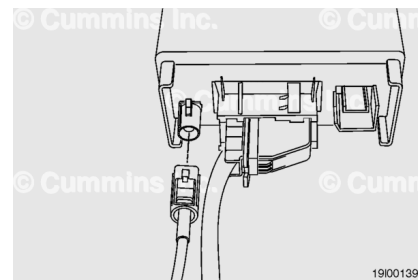
Secure the GPS antenna cable with zip ties.

Note : Do not cut the GPS antenna cable.

Connect the blue GPS antenna connector to the blue plug on the route parameter manager module.

Make sure it has fully snapped into place.

Install all cab decorative covers. See equipment manufacturer service information.



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Connect the batteries. See equipment manufacturer service information.